Contextualized Representation

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آکادمی‌راه

Outline

- **Introduction**
- ELMO
- BERT
- More from BERT Family
- GPT

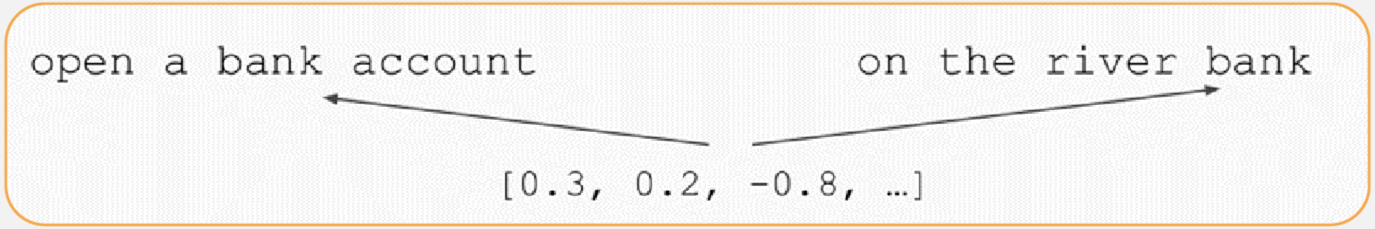
Pre-training in NLP

- Word embeddings are the basis of deep learning for NLP

- Word embeddings (word2vec, GloVe) are often pre-trained on text corpus from co-occurrence statistics

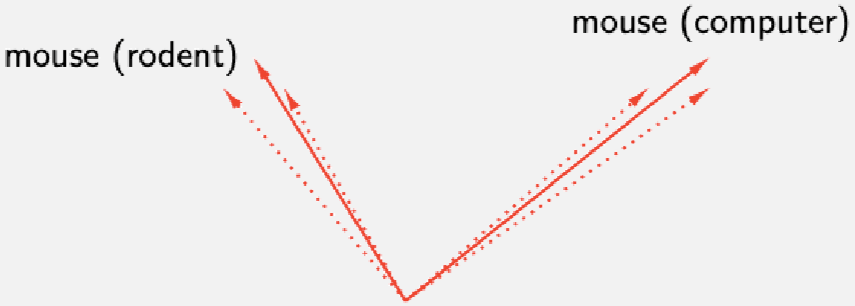
Contextual Representations

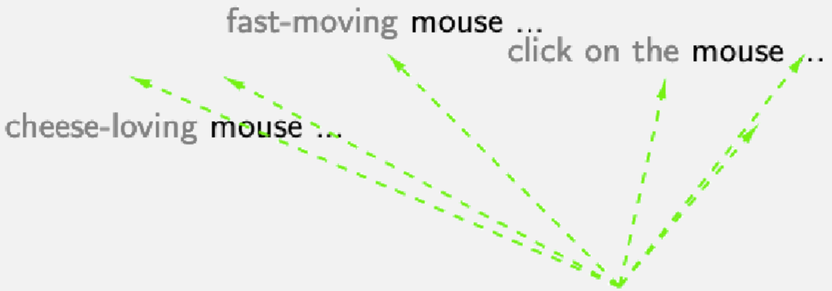
- **Problem: Word embeddings are applied in a context free manner**



- **Solution: Train contextual representations on text corpus**

Static vs Contextualized Representation





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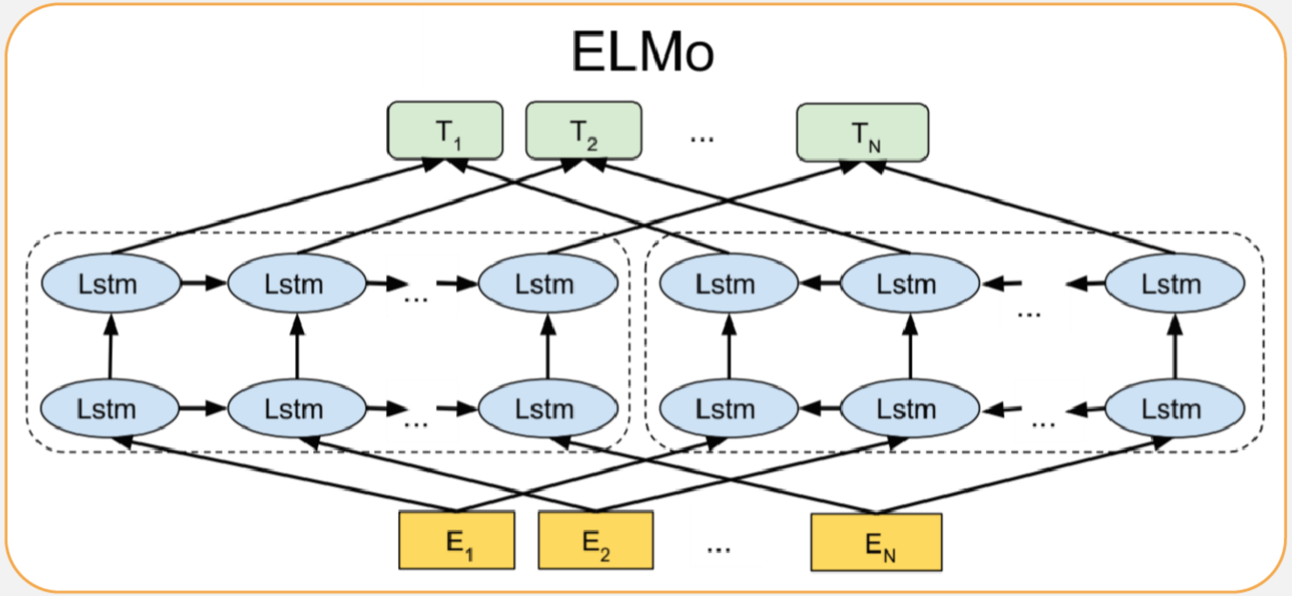
ELMO

- Bidirectional language model
- Train Separate Left-to-Right and Right-to-Left LMs



ELMO

- Bidirectional training

- Forward language model

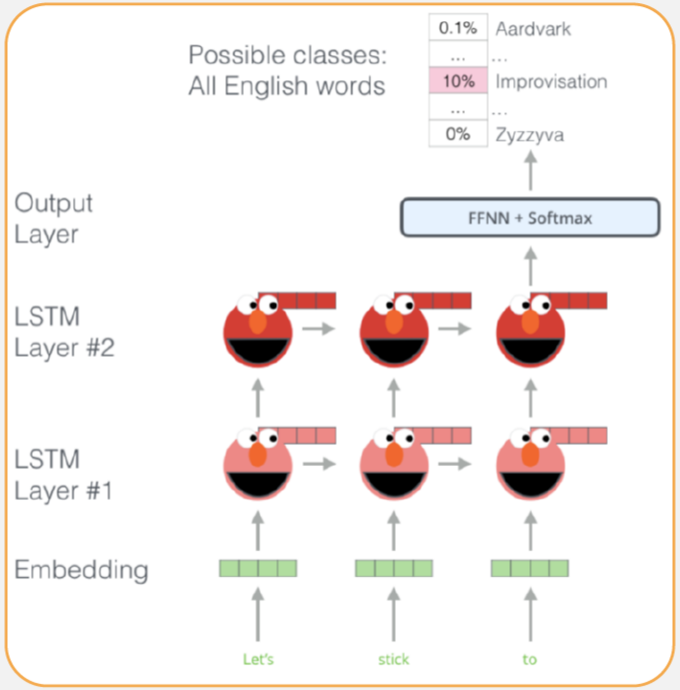
$$p(t_1, t_2; , \ldots , t_N) = \prod_{k=1}^{N} p(t_k | t_1, t_2; , \ldots , t_{k-1})$$

- Backward language model

$$p(t_1, t_2; , \ldots , t_N) = \prod_{k=1}^{N} p(t_k | t_{k+1}, t_{k+2}; , \ldots , t_N)$$

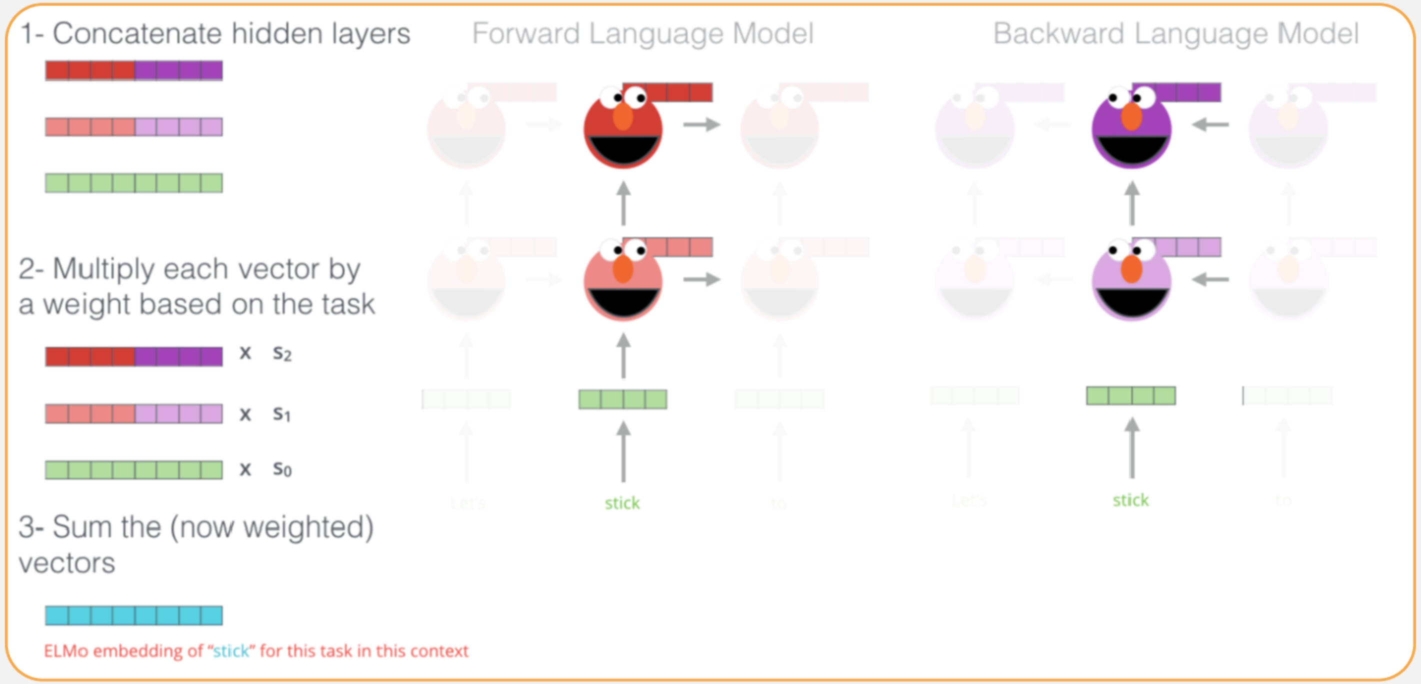
ELMO

- Prediction next word in forward training
- Prediction previous word in backward training



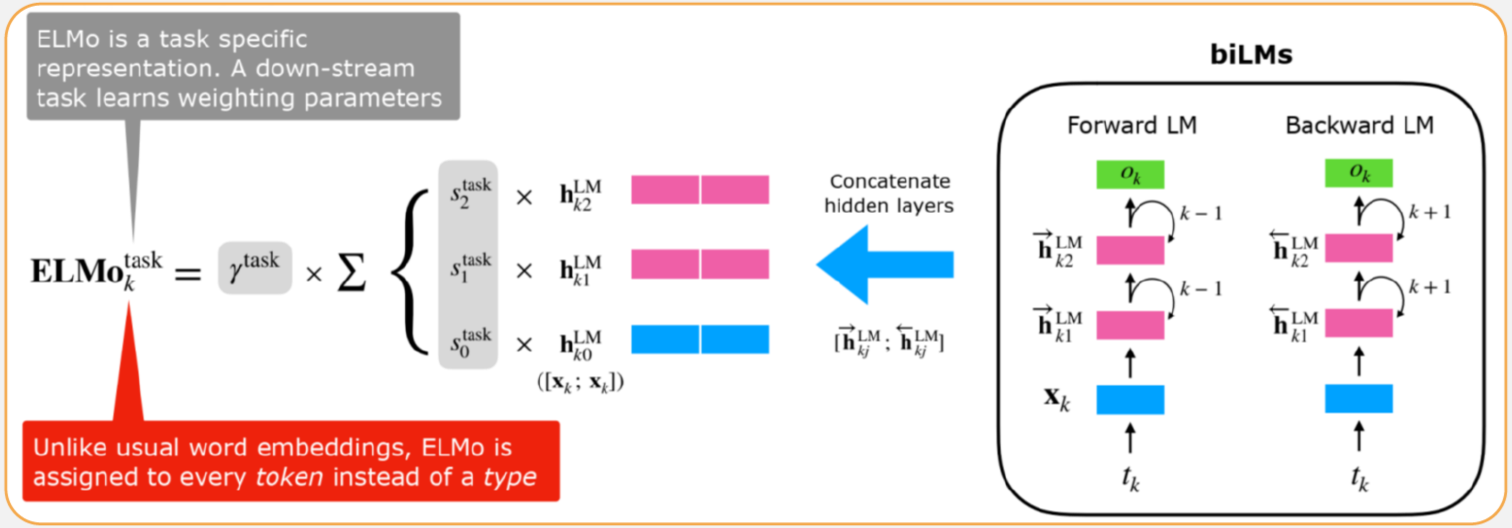
ELMO

Extracting embedding



ELMO

Extracting embedding



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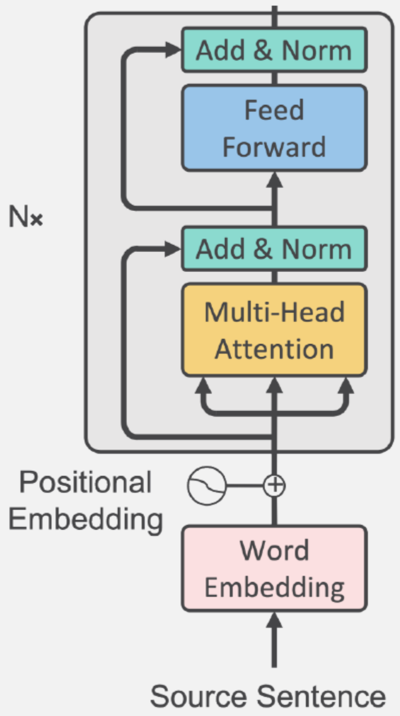
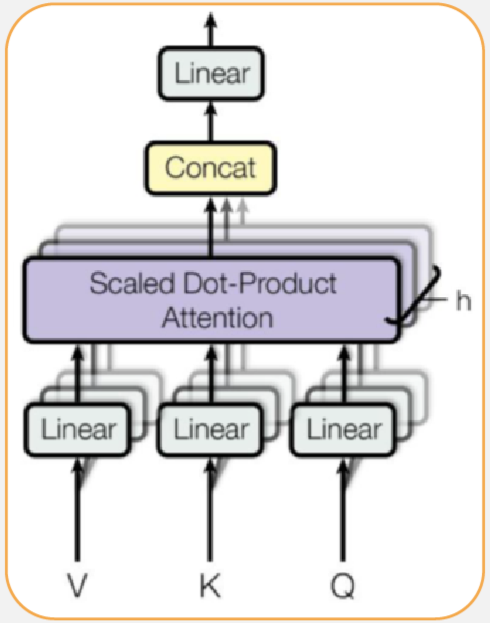
Model Architecture

- **Multi-headed self attention**
  - Models context
- **Feed-forward layers**
  - Computes non-linear hierarchical features
- **Layer norm and residuals**
  - Makes training deep networks healthy
- **Positional embeddings**
  - Allows model to learn relative positioning





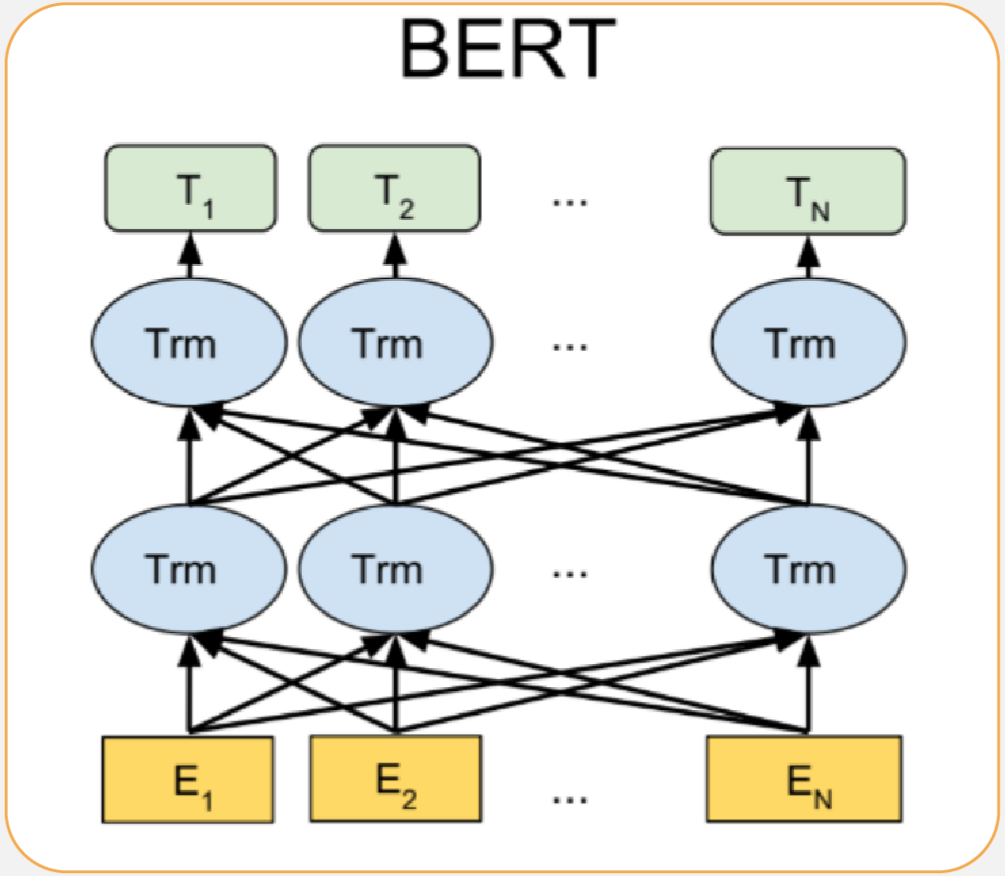
Model Architecture

- Empirical advantages of Transformer vs. LSTM
  - Self-attention == no locality bias
    - Long-distance context has "equal opportunity"
  - Single multiplication per layer == efficiency on TPU
    - Effective batch size is number of words, not sequences

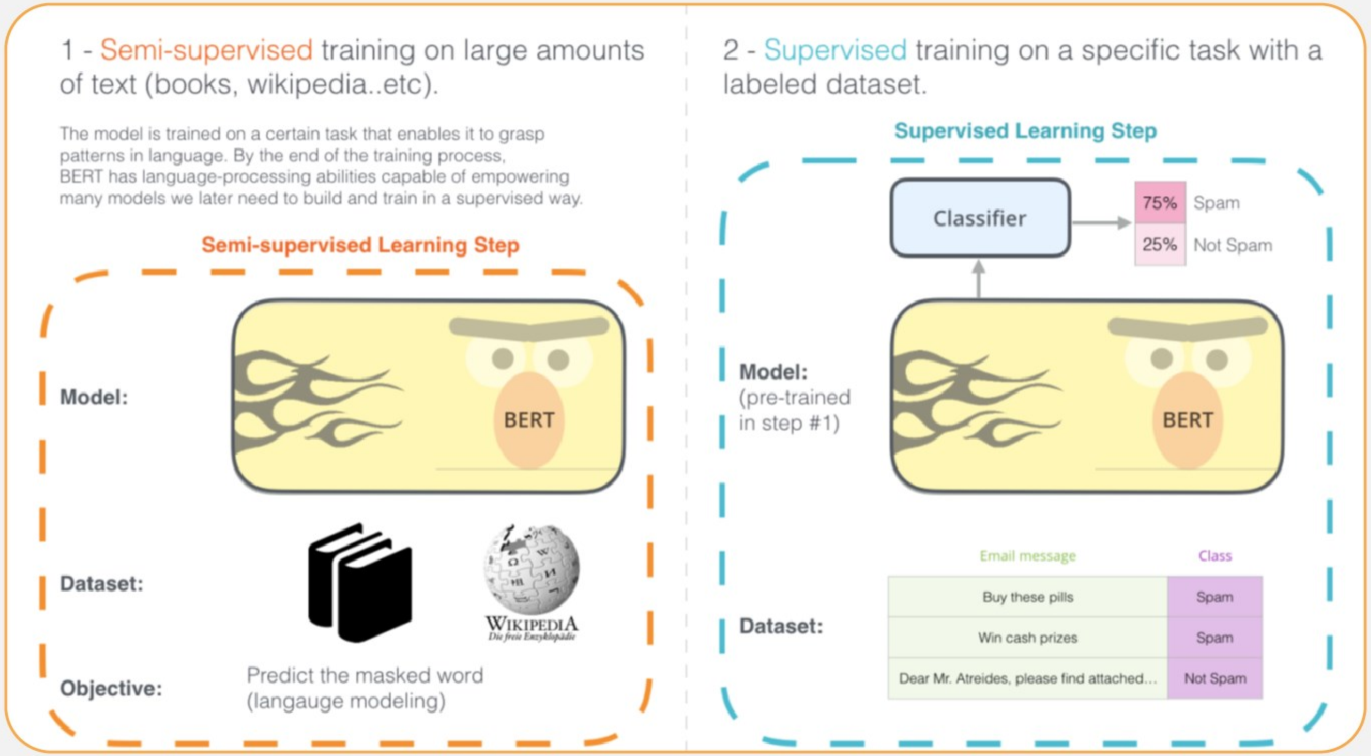
Problem with Previous Methods

- **Problem**: Language models only use left context or right context, but language understanding is bidirectional.
- Why are LMs unidirectional?
- Reason 1: Directionality is needed to generate a well-formed probability distribution.
  - We don't care about this.
- Reason 2: Words can "see themselves" in a bidirectional encoder.

BERT

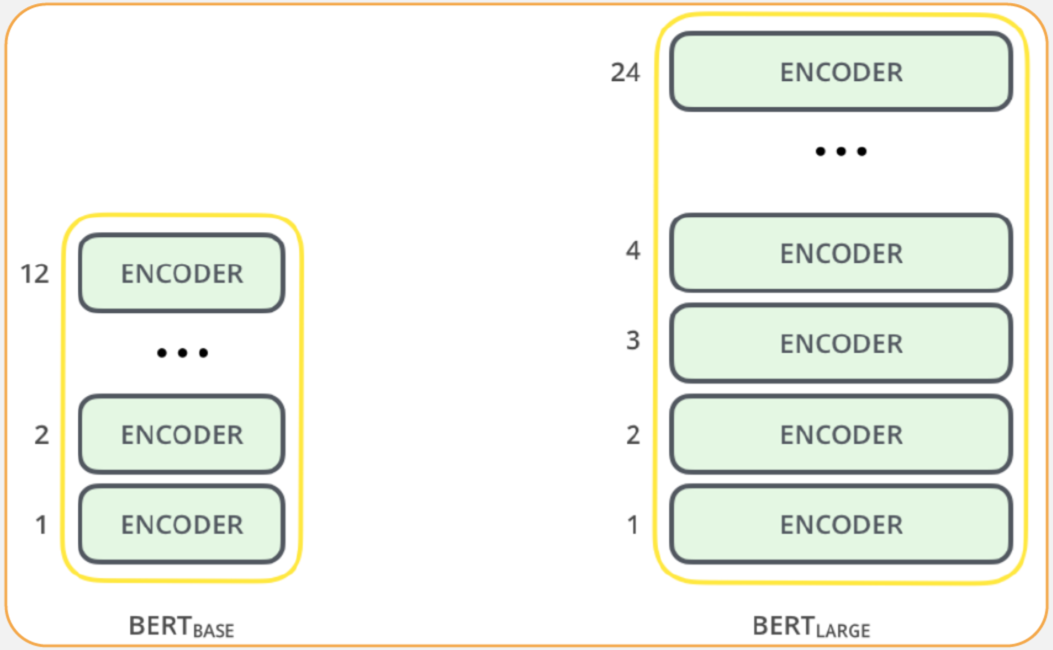


BERT


1 - Semi-supervised training on large amounts of text (books, wikipedia..etc).
The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.
Semi-supervised Learning Step
Model:
BERT
Dataset:
WIKIPEDIA
Objective:
Predict the masked word (langauge modeling)
2 - Supervised training on a specific task with a labeled dataset.
Supervised Learning Step
Classifier
75% Spam
25% Not Spam
Model: (pre-trained in step #1)
BERT
Dataset:
Email message
Class
Buy these pills
Spam
Win cash prizes
Spam
Dear Mr. Atreides, please find attached...
Not Spam

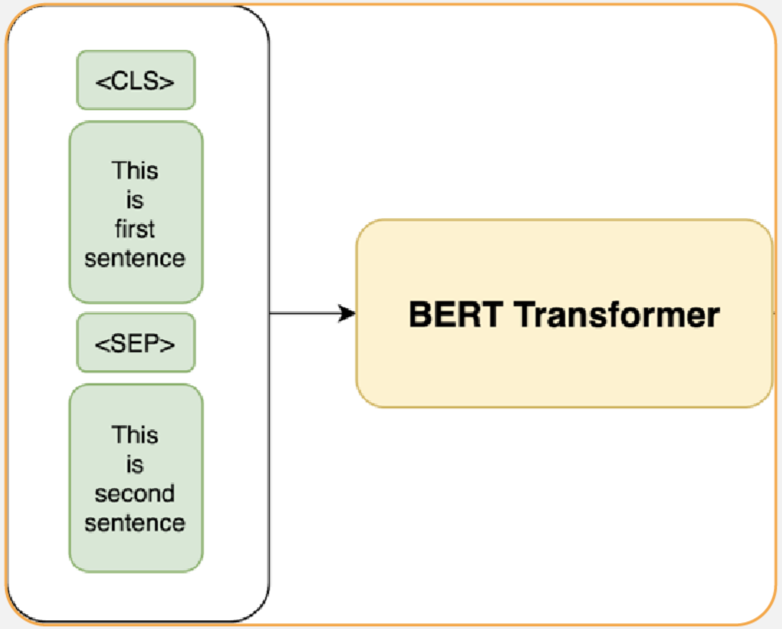
BERT

- BERT Base
- BERT Large

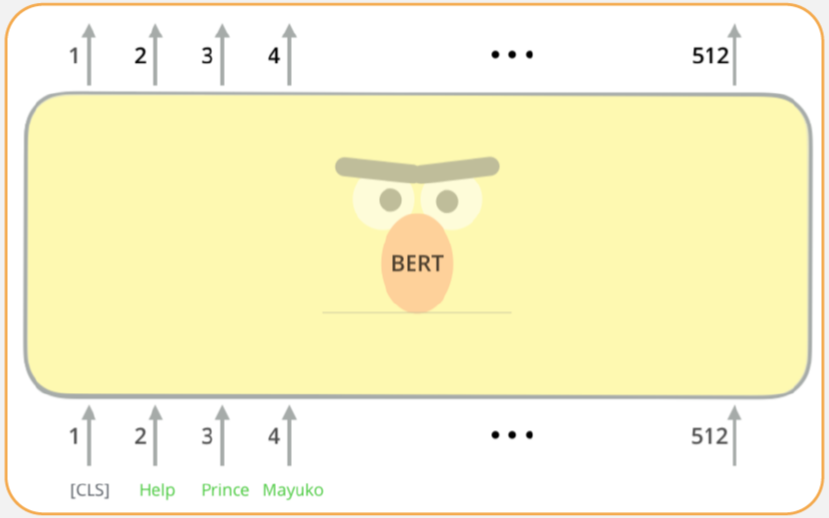


BERT

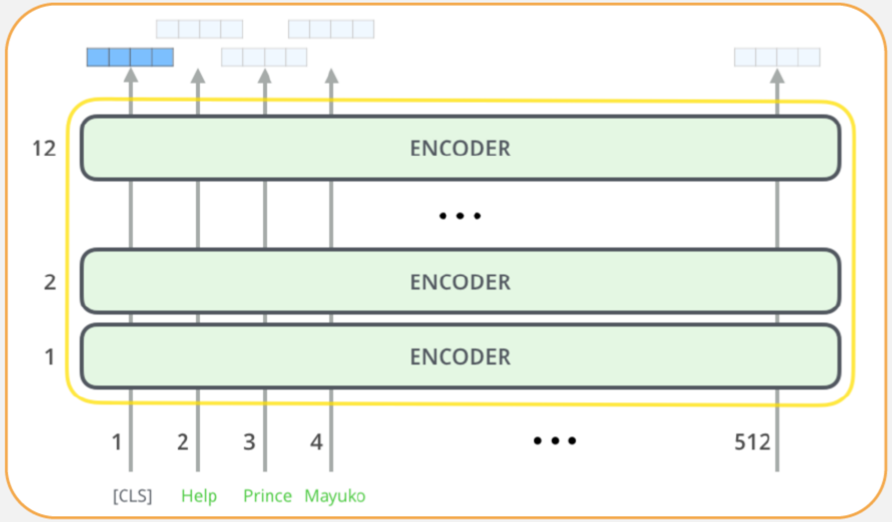
- Training type
  - Masked language model
  - Next sentence prediction
- Input
  - First token [CLS]
  - Delimiter token [SEP]
  - Masked token [MASK]



BERT



1
2
3
4
• • •
512
BERT
1
2
3
4
• • •
512
[CLS]
Help
Prince
Mayuko



12
ENCODER
• • •
2
ENCODER
1
ENCODER
1
2
3
4
• • •
512
[CLS]
Help
Prince
Mayuko

BERT

- Masked language model
- **Solution**: Mask out k% of the input words, and then predict the masked words
  - We always use k = 15%

```
                                store              gallon
                                  ↑                  ↑
the man went to the [MASK] to buy a [MASK] of milk
```

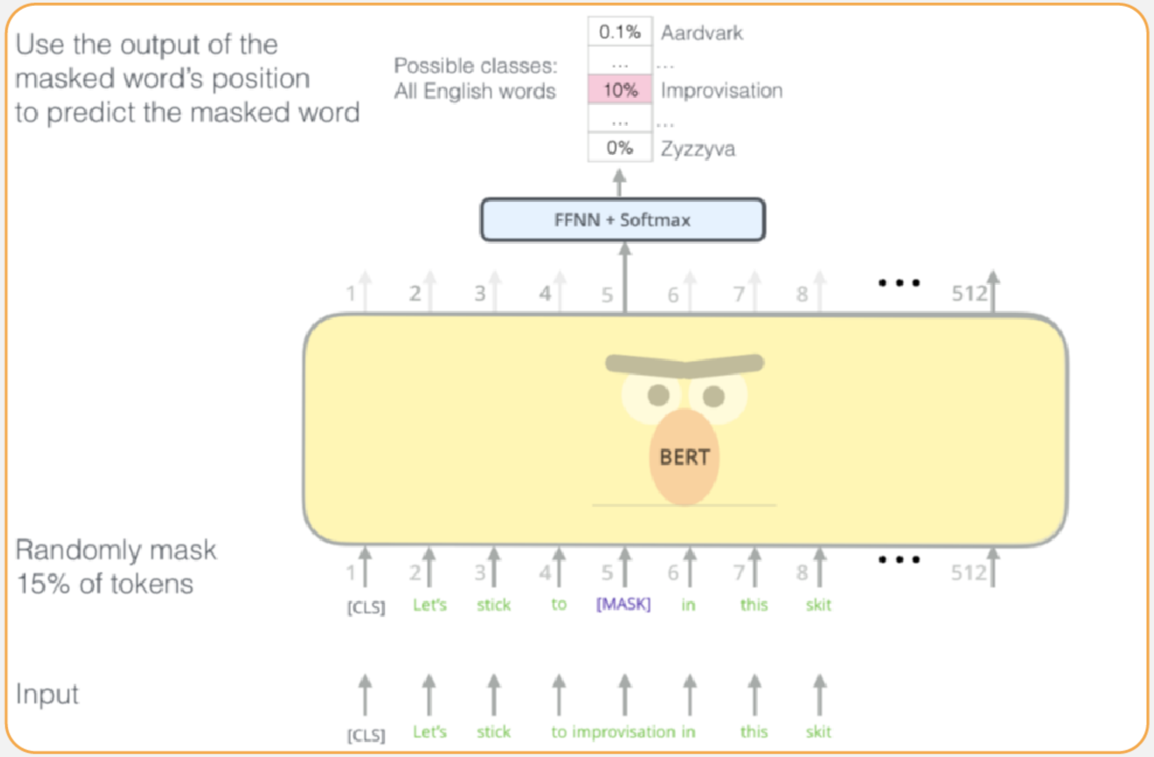
- Too little masking: Too expensive to train
- Too much masking: Not enough context

BERT

- Problem: Mask token never seen at fine-tuning
  Solution: 15% of the words to predict, but don't replace with [MASK] 100% of the time. Instead:
  - 80% of the time, replace with [MASK]
    - went to the store → went to the [MASK]
  - 10% of the time, replace random word
    - went to the store → went to the running
  - 10% of the time, keep same
    - went to the store → went to the store

BERT

Masked language model

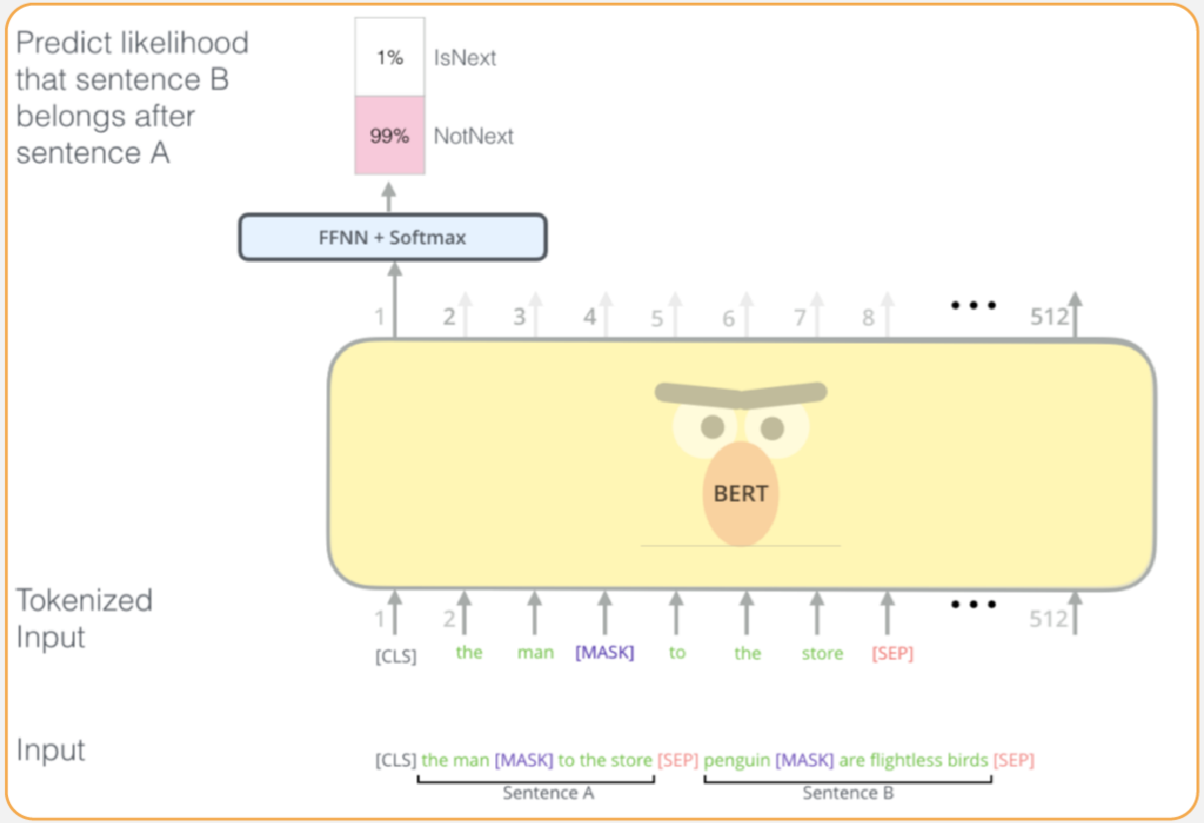


BERT

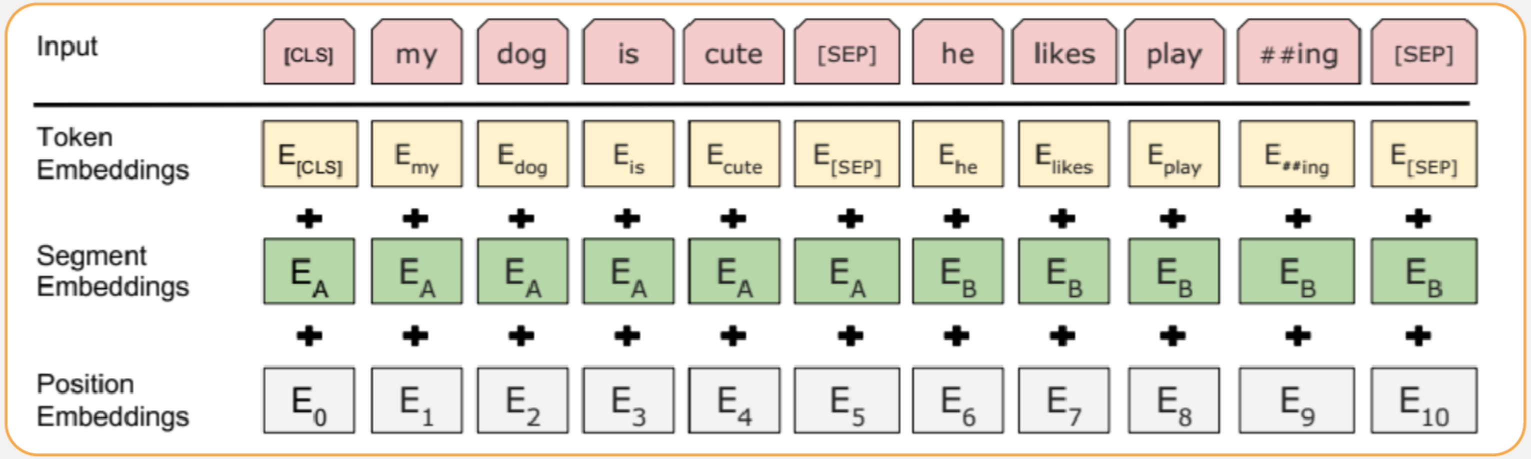
- Next sentence prediction
- To learn relationships between sentences, predict whether Sentence B is actual sentence that proceeds Sentence A, or a random sentence

Sentence A = `The man went to the store.`	**Sentence A** = `The man went to the store.`
Sentence B = `He bought a gallon of milk.`	**Sentence B** = `Penguins are flightless.`
Label = `IsNextSentence`	**Label** = `NotNextSentence`

Next sentence prediction

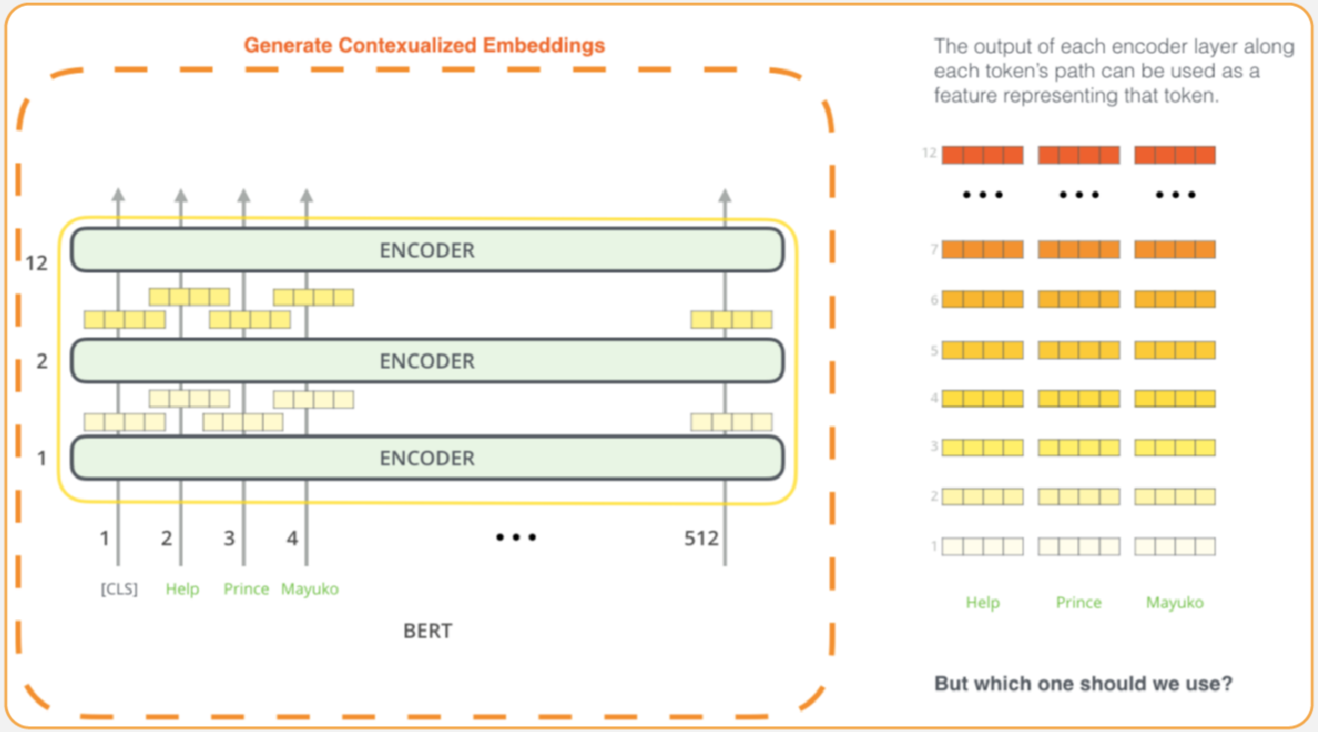


BERT


Input
[CLS]
my
dog
is
cute
[SEP]
he
likes
play
##ing
[SEP]
Token Embeddings
Segment Embeddings
Position Embeddings

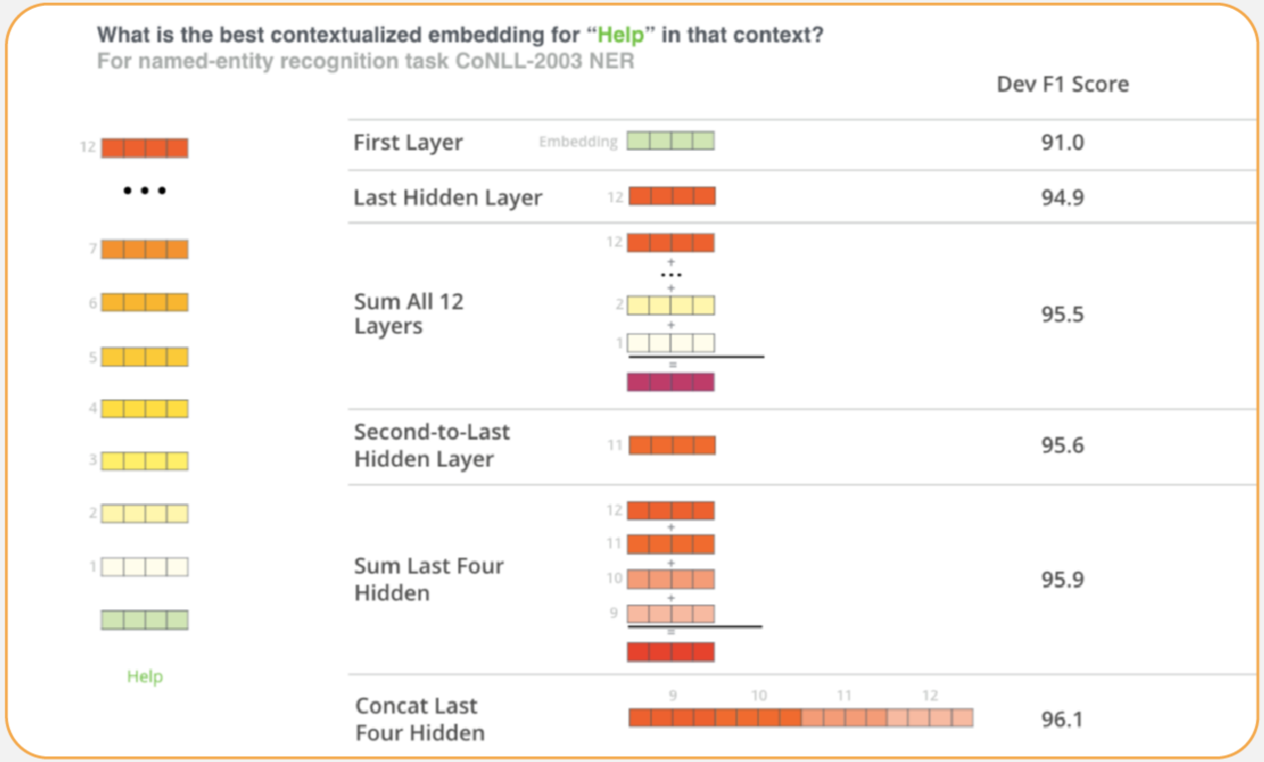
BERT

Extracting embedding

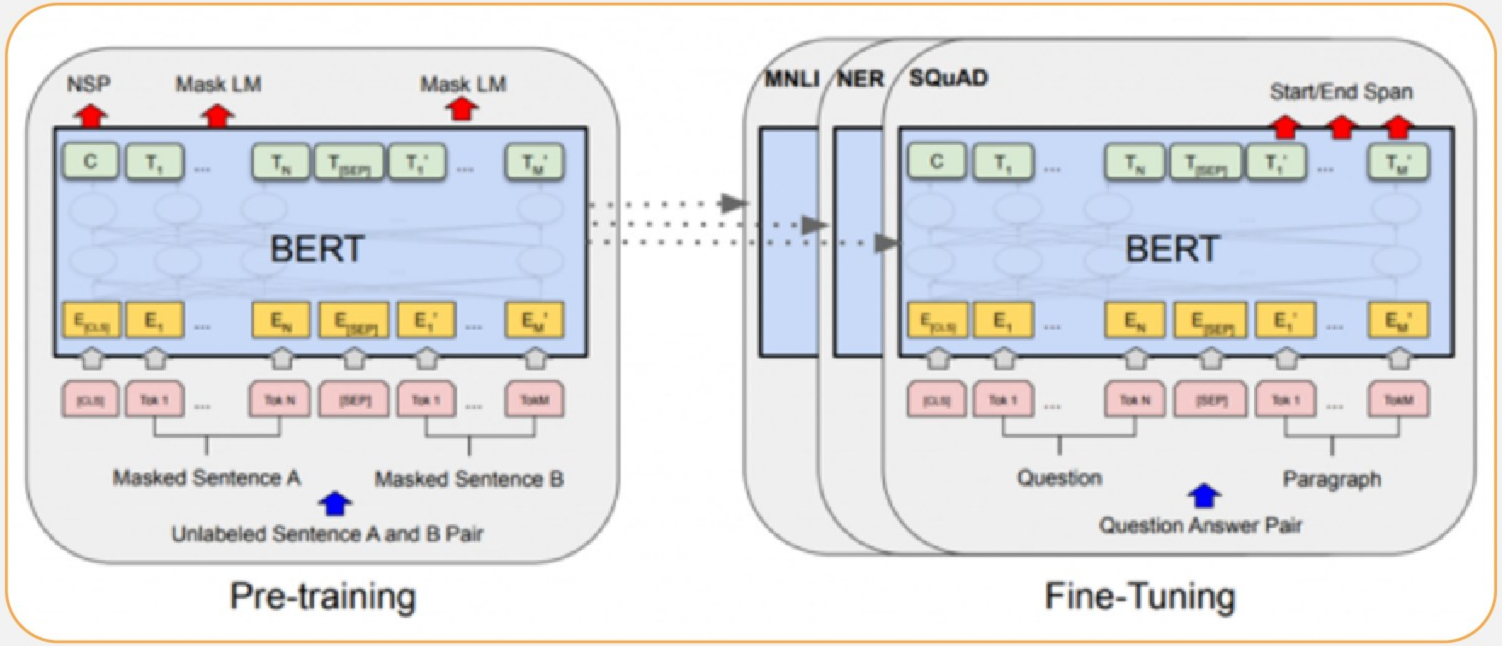


BERT

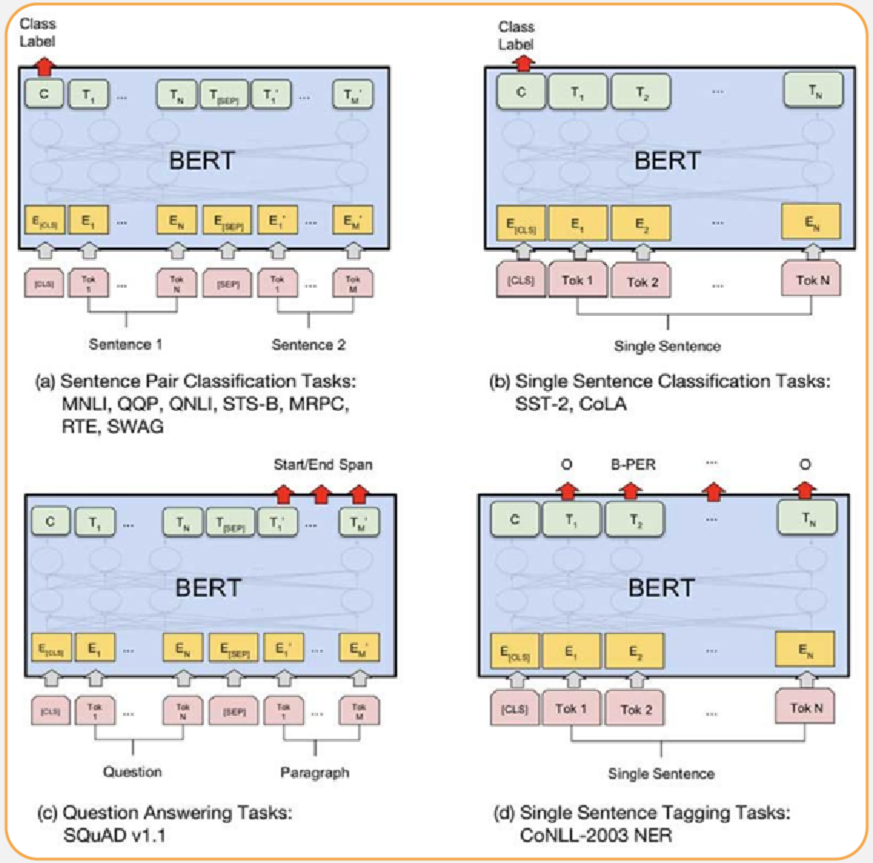
Extracting embedding



Fine-Tuning Procedure



Fine-Tuning Procedure



Motivations for Improving BERT

- Accuracy
- Large
- Slow
- Hard to train

	Bert
Size(millions)	Base: 110 Large: 340
Training Time	Base: 8 x v100 days* Large:64 TPU Chips x 4 days (or 280 x V100 x 1 days)
Performance	outperforms state of the art in Oct 2018
Date	16 GB BERT data (Book Corpus + Wikipedia). 3.3 Billion words.
Method	BERT (Bidirectional Transformer with MLM and NSP)

Model Improvement Methods

- Larger with more data
- Quantization
  - Quantization-aware training
- Distillation
- Pruning
- Specialization

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More from BERT Family

- RoBERTa
- XLNET
- ALBERT
- T5
- ELECTRA
- DistilBERT

RoBERTa

- RoBERTa: A Robustly Optimized BERT Pretraining Approach
  - Trained BERT for more epochs and/or on more data
    - Showed that more epochs alone helps, even on same data
    - More data also helps
  - Next Sentence Prediction (NSP) removed
  - Dynamic Masking

	MNLI	QNLI	QQP	RTE	SST	MRPC	CoLA	STS	WNLI	Avg
Single-task single models on dev										
$\mathrm{BERT}_{\mathrm{LARGE}}$	86.6/-	92.3	91.3	70.4	93.2	88.0	60.6	90.0	-	-
$\mathrm{XLNet}_{\mathrm{LARGE}}$	89.8/-	93.9	91.8	83.8	95.6	89.2	63.6	91.8	-	-
RoBERTa	**90.2/90.2**	**94.7**	**92.2**	**86.6**	**96.4**	**90.9**	**68.0**	**92.4**	**91.3**	-

(Liu et al, University of Washington and Facebook, 2019)

XLNET

- XLNet: Generalized Autoregressive Pretraining for Language Understanding (Yang et al, CMU and Google, 2019)
- More data and training time
- Permutation language modeling

XLNET

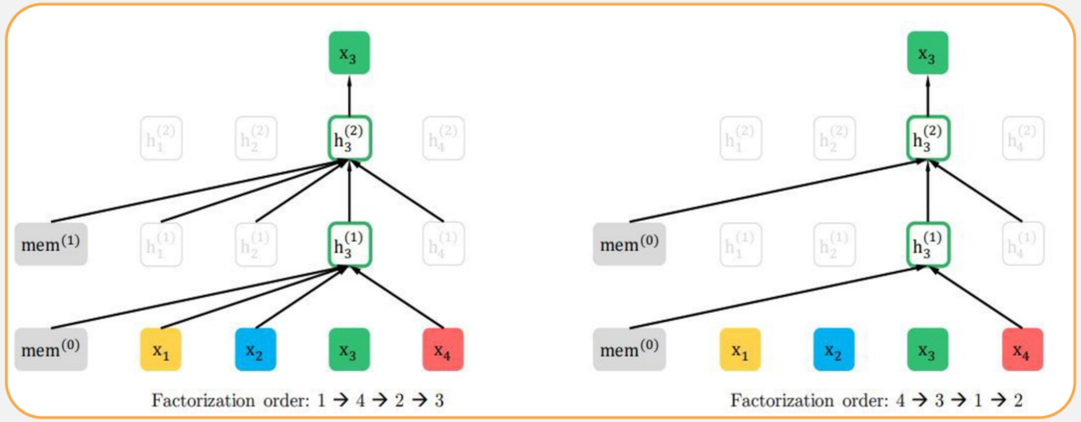
Innovation #1: Relative position embeddings

- Sentence: John ate a hot dog
- Absolute attention: "How much should dog attend to hot (in any position), and how much should dog in position 4 attend to the word in position 3? (Or 508 attend to 507, …)"
- Relative attention: "How much should dog attend to hot (in any position) and how much should dog attend to the previous word?"

XLNET

Innovation #2: Permutation Language Modeling

- In a left-to-right language model, every word is predicted based on all of the words to its left
- Instead: Randomly permute the order for every training sentence
- Equivalent to masking, but many more predictions per sentence
- Can be done efficiently with Transformers



XLNET

- Also used more data and bigger models, but showed that innovations improved on BERT even with same data and model size
- XLNet results:

Model	MNLI	QNLI	QQP	RTE	SST-2	MRPC	CoLA	STS-B
Single-task single models on dev								
BERT [2]	86.6/-	92.3	91.3	70.4	93.2	88.0	60.6	90.0
RoBERTa [21]	90.2/90.2	94.7	92.2	**86.6**	96.4	**90.9**	68.0	92.4
XLNet	**90.8/90.8**	**94.9**	**92.3**	85.9	**97.0**	90.8	**69.0**	**92.5**

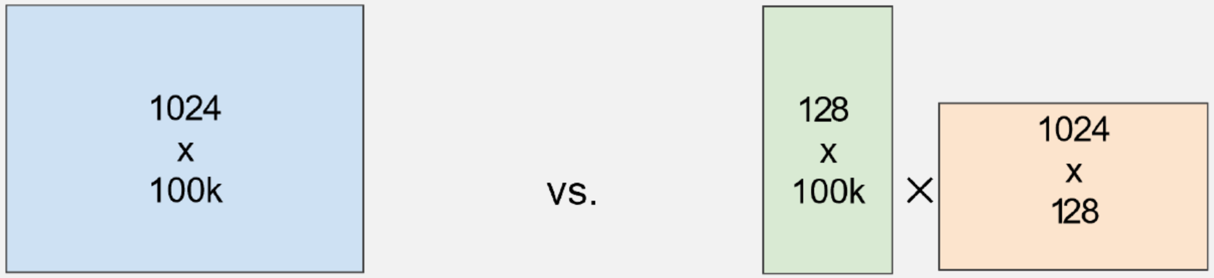
- MNLI: Multi-Genre Natural-Language-Inference

ALBERT

- ALBERT: A Lite BERT for Self-supervised Learning of Language Representations
- Factorization of embedding layers
- Cross-layer parameter sharing
  - (Lan et al, Google and TTI Chicago, 2019)

ALBERT

- Innovation #1: Factorized embedding parametrization
- Break down token embeddings into two small embedding matrixes
- After applying this decomposition, embeddings parameters can be reduced
  - from (number of tokens * hidden layer size)
  - to (number of tokens * token embedding size + token embedding size * hidden layer size)



ALBERT

Innovation #2: Cross-layer parameter sharing

- Share all parameters between Transformer layers
- Results:

Models	MNLI	QNLI	QQP	RTE	SST	MRPC	CoLA	STS
Single-task single models on dev								
BERT-large	86.6	92.3	91.3	70.4	93.2	88.0	60.6	90.0
XLNet-large	89.8	93.9	91.8	83.8	95.6	89.2	63.6	91.8
RoBERTa-large	90.2	94.7	**92.2**	86.6	96.4	**90.9**	68.0	92.4
ALBERT (1M)	90.4	95.2	92.0	88.1	96.8	90.2	68.7	92.7
ALBERT (1.5M)	**90.8**	**95.3**	**92.2**	**89.2**	**96.9**	**90.9**	**71.4**	**93.0**

ALBERT is light in terms of parameters, not speed

Model		Parameters	SQuAD1.1	SQuAD2.0	MNLI	SST-2	RACE	Avg	Speedup
BERT	base	108M	90.4/83.2	80.4/77.6	84.5	92.8	68.2	82.3	4.7x
	large	334M	92.2/85.5	85.0/82.2	86.6	93.0	73.9	85.2	1.0
ALBERT	base	12M	89.3/82.3	80.0/77.1	81.6	90.3	64.0	80.1	5.6x
	large	18M	90.6/83.9	82.3/79.4	83.5	91.7	68.5	82.4	1.7x
	xlarge	60M	92.5/86.1	86.1/83.1	86.4	92.4	74.8	85.5	0.6x
	xxlarge	235M	**94.1/88.3**	**88.1/85.1**	**88.0**	**95.2**	**82.3**	**88.7**	0.3x

Real Application

- Applying Models
- BERT and other pre-trained language models are extremely large and expensive
- How are companies applying them to low-latency production services?

GOOGLE \ TECH \ ARTIFICIAL INTELLIGENCE

Google is improving 10 percent of searches by understanding language context

Say hello to BERT

By Dieter Bohn | @backlon | Oct 25, 2019, 3:01am EDT

Bing says it has been applying BERT since April

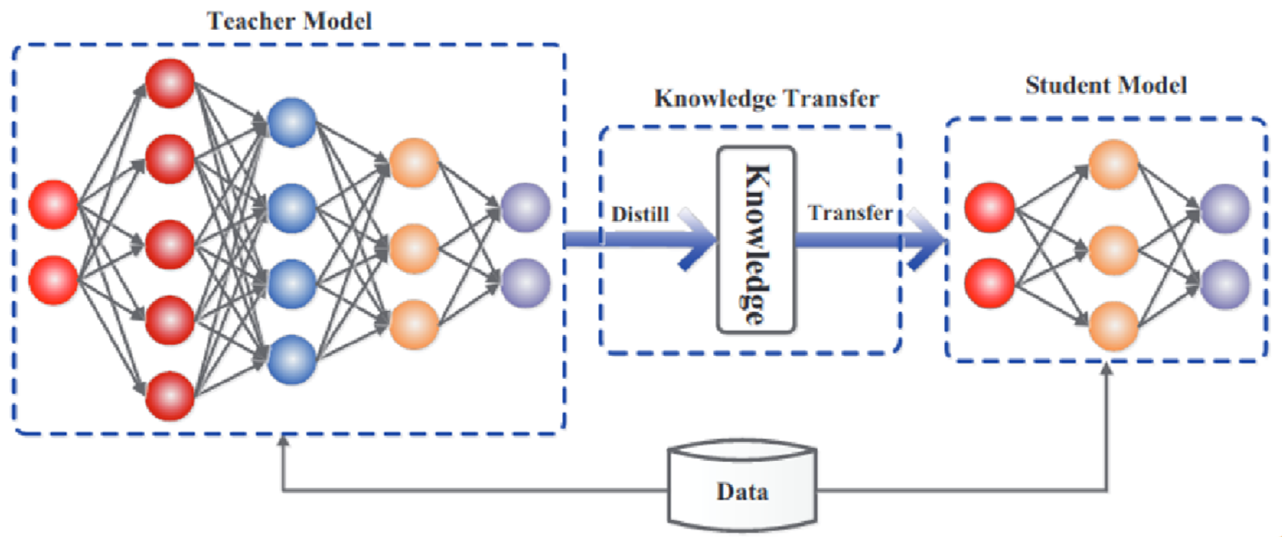
The natural language processing capabilities are now applied to all Bing queries globally.

George Nguyen on November 19, 2019 at 1:38 pm

Distillation

- Answer: Distillation (a.k.a., model compression)
- Idea has been around for a long time:
  - Model Compression (Bucila et al, 2006)
  - Distilling the Knowledge in a Neural Network (Hinton et al, 2015)
- Simple technique
  - Train "Teacher": Use SOTA pre-training + fine-tuning technique to train model with maximum accuracy
  - Label a large amount of unlabeled input examples with Teacher
  - Train "Student": Much smaller model (e.g., 50x smaller) which is trained to mimic Teacher output
  - Student objective is typically Mean Square Error or Cross Entropy

Distillation



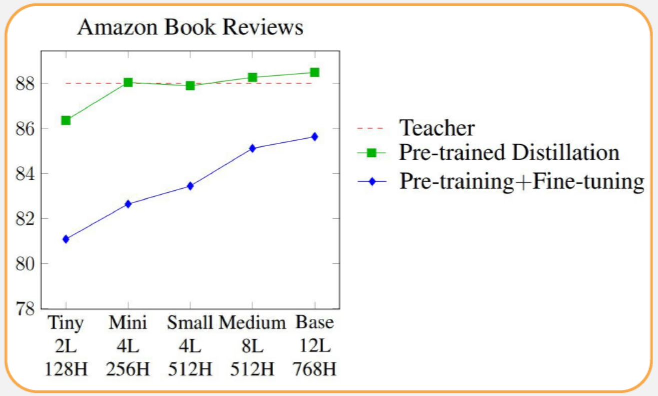
DistilBERT

- Faster inference
- Faster training

Comparison	BERT October 11, 2018	DistilBERT October 2, 2019
Parameters	**Base:** 110M **Large:** 340M	**Base:** 66
Layers / Hidden Dimensions / Self-Attention Heads	**Base:** 12 / 768 / 12 **Large:** 24 / 1024 / 16	**Base:** 6 / 768 / 12
Training Time	**Base:** 8 x V100 x 12d **Large:** 280 x V100 x 1d	**Base:** 8 x V100 x 3.5d (4 times less than BERT)
Performance	Outperforming SOTA in Oct 2018	97% of BERT-base's performance on GLUE
Pre-Training Data	BooksCorpus + English Wikipedia = 16 GB	BooksCorpus + English Wikipedia = 16 GB
Method	Bidirectional Trans-former, MLM & NSP	BERT Distillation

Distillation

- Example distillation results
  - 50k labeled examples, 8M unlabeled examples



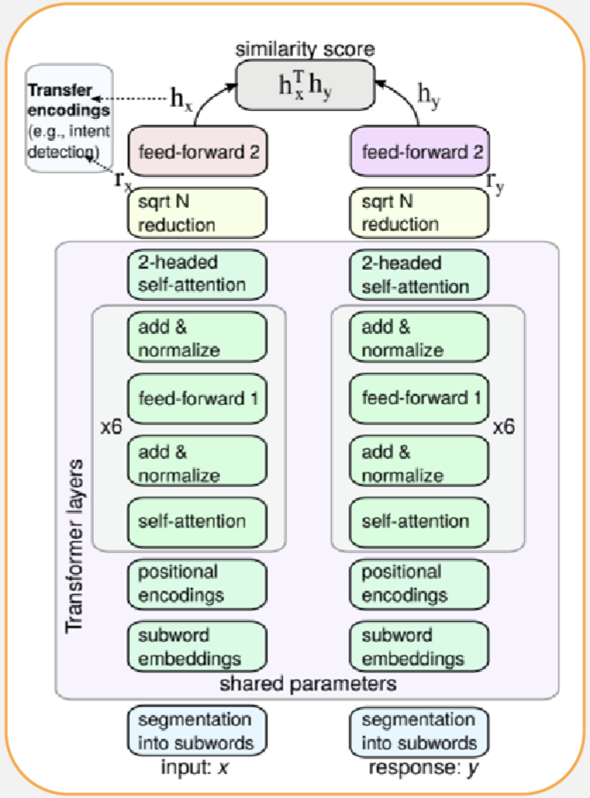
  - Well-Read Students Learn Better: On the Importance of Pre-training Compact Models (Turc et al, 2020)
- Distillation works much better than pre-training + fine-tuning with smaller model

Why does distillation work so well?

- Fine-tuning mostly just picks up and tweaks existing latent features
- This requires an oversized model, because only a subset of the features are useful for any given task
- Distillation allows the model to only focus on those features
- Supporting evidence: Simple self-distillation of a small model (e.g., distilling a smaller BERT model) doesn't work very well

Specialization

- Conversational Representations from Transformers (ConveRT)
  - Less complex model
  - Specialized training data



Comparison

	BERT	RoBERTa	DistilBERT	XLNet
Size (millions)	**Base**: 110 **Large**: 340	**Base**: 110 **Large**: 340	**Base**: 66	**Base**: ~110 **Large**: ~340
Training Time	**Base**: 8 x V100 x 12 days* **Large**: 64 TPU Chips x 4 days (or 280 x V100 x 1 days*)	**Large**: 1024 x V100 x 1 day; 4-5 times more than BERT.	**Base**: 8 x V100 x 3.5 days; 4 times less than BERT.	**Large**: 512 TPU Chips x 2.5 days; 5 times more than BERT.
Performance	Outperforms state-of-the-art in Oct 2018	2-20% improvement over BERT	3% degradation from BERT	2-15% improvement over BERT
Data	16 GB BERT data (Books Corpus + Wikipedia). 3.3 Billion words.	160 GB (16 GB BERT data + 144 GB additional)	16 GB BERT data. 3.3 Billion words.	**Base**: 16 GB BERT data **Large**: 113 GB (16 GB BERT data + 97 GB additional). 33 Billion words.
Method	BERT (Bidirectional Transformer with MLM and NSP)	BERT without NSP**	BERT Distillation	Bidirectional Transformer with Permutation based modeling

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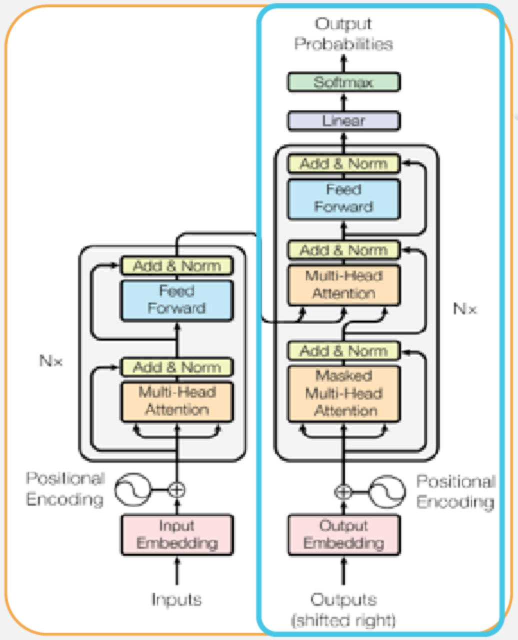
Generative Pre-Training (GPT)

Elmo
(94M)

Bert
(340M)

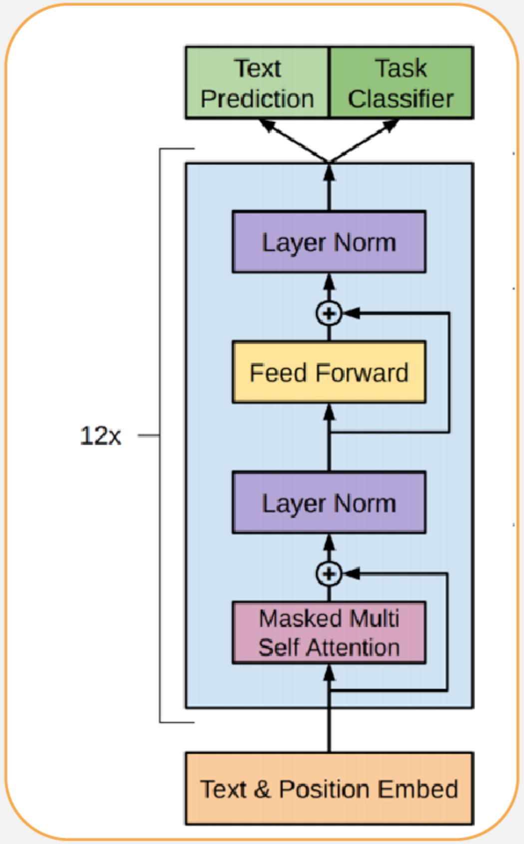


Transformer Decoder

GPT-2
(1.5B)

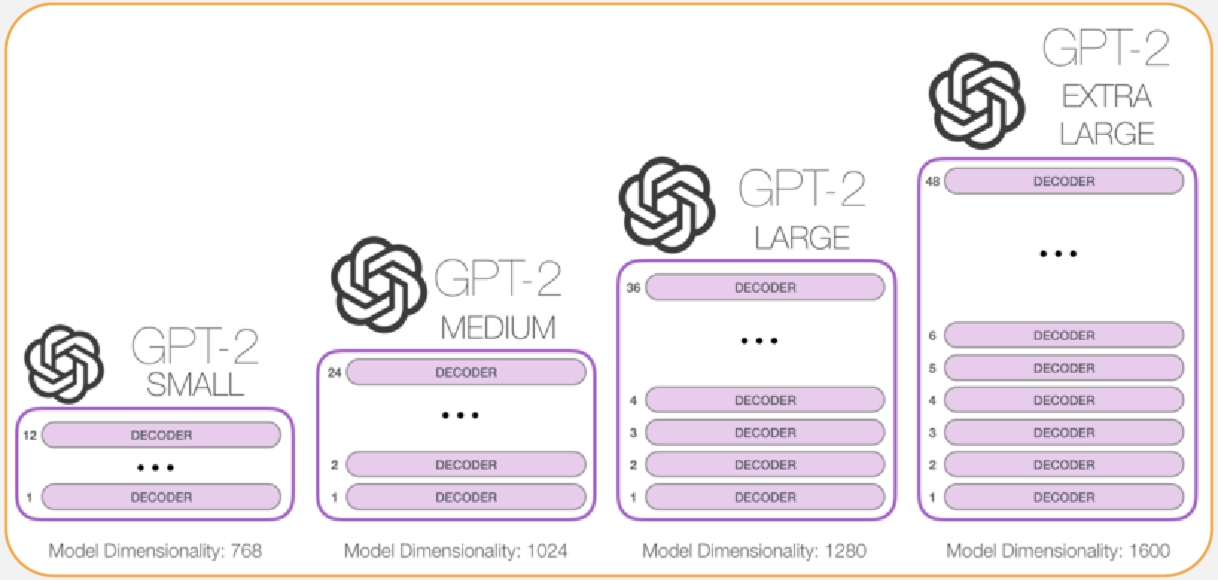
GPT-1

- 12 decoders architecture
- Semi-supervised training
- Zero-shot and few-shot learning



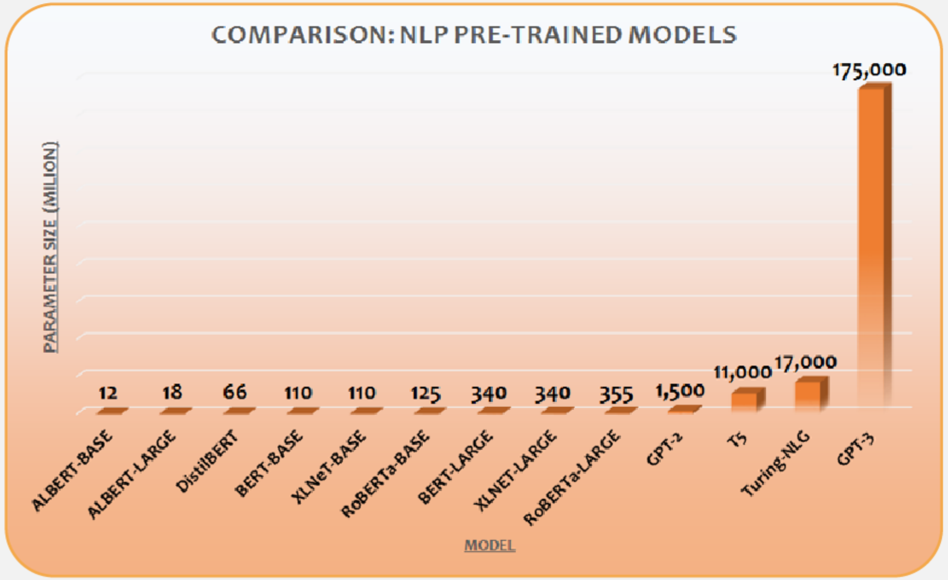
GPT-2

- 48 layer
- Larger with more data (1.5 billion vs 117 million)
- Task conditioning



GPT-3

- 96 layers
- Much larger than GPT-2
- Not accessible



Further Reading

- Speech and Language Processing (3rd ed. draft)
  - Chapter 10 (not released yet)

References

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- https://mapmeld.medium.com/a-whole-world-of-bert-f20d6bd47b2f
- https://ai.googleblog.com/2019/12/albert-lite-bert-for-self-supervised.html
- Yang, Zhilin, et al. "Xlnet: Generalized autoregressive pretraining for language understanding." arXiv preprint arXiv:1906.08237 (2019).

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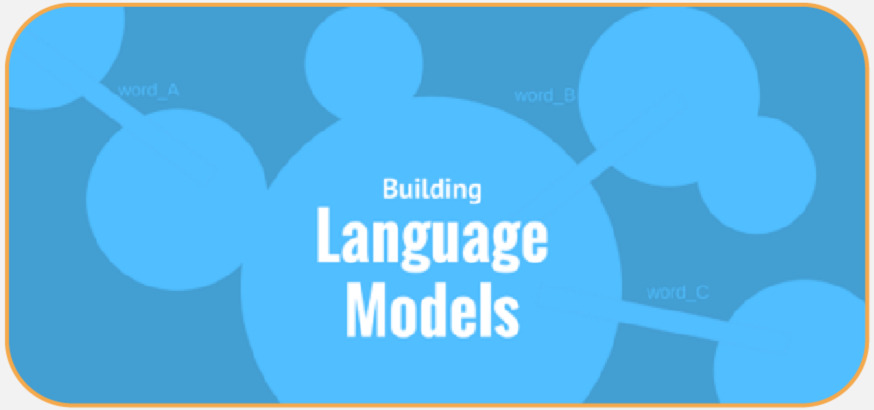
- Jiang, Zihang, et al. "Convbert: Improving bert with span-based dynamic convolution." arXiv preprint arXiv:2008.02496 (2020).
- https://blog.rasa.com/how-to-benchmark-bert/

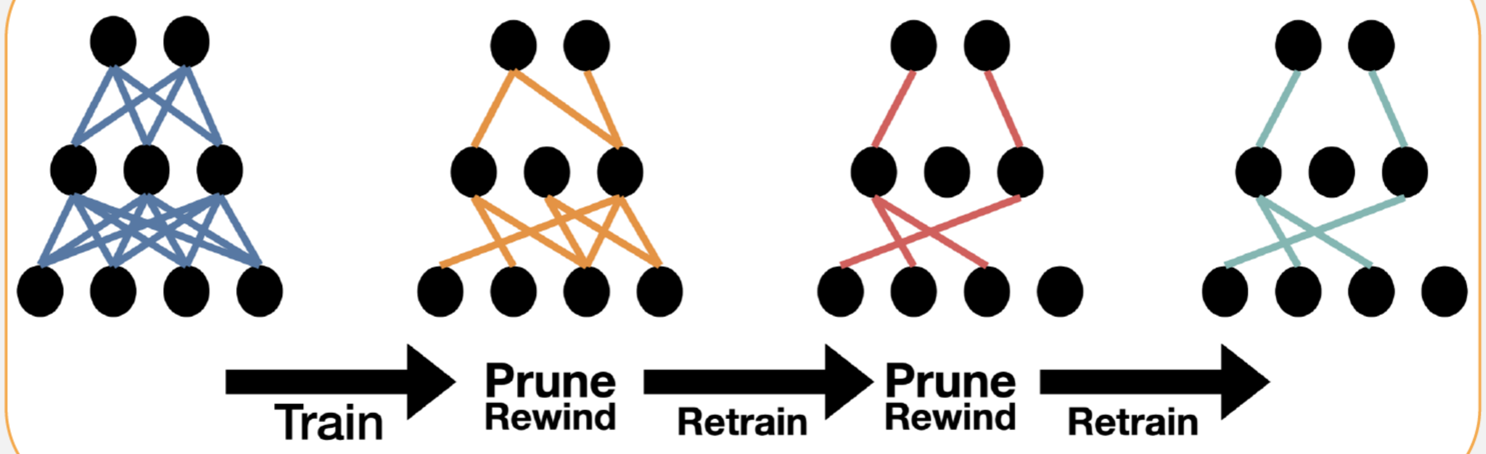
Masked language model

- مشکل: در زمان تست، توکن [MASK] وجود ندارد.
- اگر توکن i انتخاب شده بود:
  - جایگزین با [MASK] : 80%
  - جایگزین با توکن تصادفی: 10%
  - عدم تغییر توکن: 10%
- بهبود generalization شبکه

ترانسفورمر | ELMO | BERT | منابع



Pruning



mBERT

- Multi-lingual BERT
- Not as good as language specific BERT

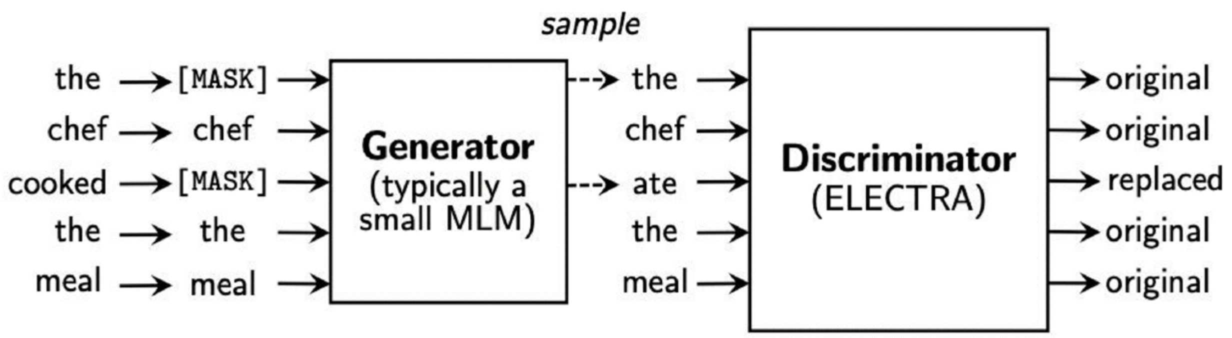
T5

- Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer
- Ablated many aspects of pre-training
  - Model size
  - Amount of training data
  - Domain/cleanness of training data
  - Pre-training objective details (e.g., span length of masked text)
  - Ensembling
  - Finetuning recipe (e.g., only allowing certain layers to finetune)
  - Multi-task training

(Raffel et al, Google, 2019)

ELECTRA

- ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators
- Train model to discriminate locally plausible text from real text
- Used, e.g., with ALBERT



(Clark et al, 2020)

One-shot or Few-shot Learning?

- Reading Comprehension
- Summarization
- Translation

Examples

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

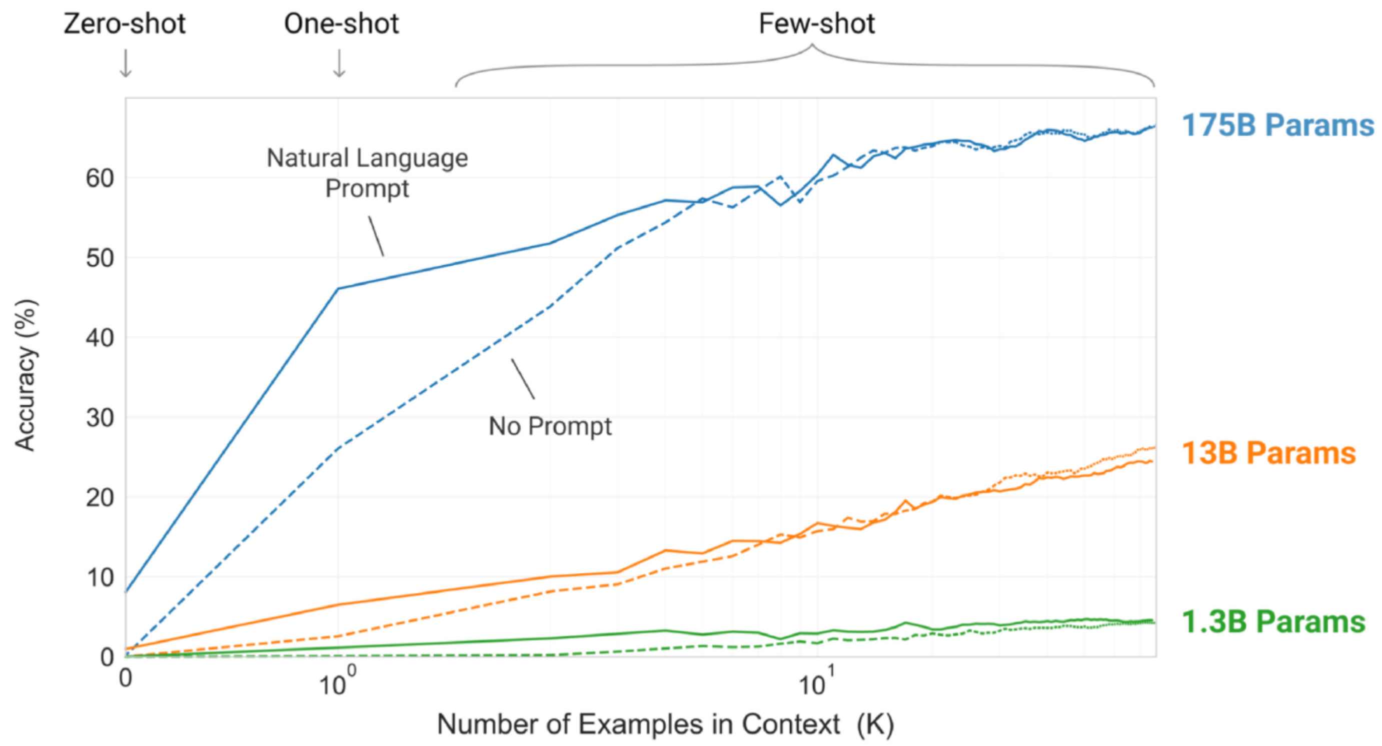
```
Translate English to French:     ← task description
sea otter => loutre de mer       ← example
cheese =>                        ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:       ← task description
sea otter => loutre de mer         ← examples
peppermint => menthe poivrée       ←
plush girafe => girafe peluche     ←
cheese =>                          ← prompt
```

Evaluation


Zero-shot
One-shot
Few-shot
Natural Language Prompt
No Prompt
Accuracy (%)
60
50
40
30
20
10
0
0
10^0
10^1
Number of Examples in Context (K)
175B Params
13B Params
1.3B Params